

Autodesk® Scaleform®

Scaleform 3.3

: Michael Antonov, Maxim Shemanarev
: 4.01
: 2011 2 17

Copyright Notice

Autodesk® Scaleform® 4.2

© 2012 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AUGI, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo) Built with ObjectARX (design/logo), Burn, Buzzsaw, CAiCE, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, DesignServer, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWFX, DXF, Ecotect, Evolver, Exposure, Extending the Design Team, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, Freewheel, GDX Driver, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, Tinkerbox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR

IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

:

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

목차

1	Scaleform 3.x 메모리 시스템	1
1.1		1
1.1.1		1
1.1.2		2
1.1.3		3
1.1.4		3
1.2	Scaleform 3.3 API	4
2	메모리 할당	5
2.1		6
2.1.1	SysAlloc	6
2.2	Scaleform	7
2.2.1	(Memory Arenas)	8
2.3	OS	10
2.3.1	SysAllocPaged	11
2.4		14
2.4.1	API	14
2.4.2		16
3	메모리 보고	20
4	가베지 콜렉션	25
4.1		26

4.2	Gfx::Movie:: ForceCollectGarbage	28
-----	--	----

1 Scaleform 3.x 메모리 시스템

Autodesk® Scaleform® 3.0 (heap) ,
Scaleform 3.2
AMP(Analyzer for Memory and Performance)
Scaleform 3.3
1.2
Scaleform 3.3
Movies Scaleform
ActionScript (Garbage Collection)

1.1 중요한 메모리 개념

Scaleform / Scaleform

1.1.1 오버라이드 할당자

Scaleform Gfx::System

1. Scaleform::SysAlloc Alloc/Free/Realloc
dImalloc

2. Scaleform

3. Scaleform (: Scaleform::SysAllocWinAPI)

(2)

2 '

1.1.2 메모리 힙

Scaleform

AMP MemReport

Global -

MovieData - SWF/GFx

MovieView - GFx::Movie ActionScript

MeshCashe - shape triangle data

Scaleform 3.3

Scaleform::SysAlloc

, Scaleform 3.3

GfX::Movie

1.1.3 조각모음

Scaleform 3.0 AS(ActionScript)

AS

CLIK

ActionScript

Scaleform GfX::Movie

, AS

. Scaleform 3.3 Movie

(garbage collector)

ActionScript

Movie

Scaleform

4:

1.1.4 메모리 보고 및 디버깅

Scaleform "stat ID"

stat ID

stat ID

Scaleform 3.2

AMP(Analyzer for Memory and Performance)

. AMP

[AMP](#)

, Scaleform::MemoryHeap::MemReport

stat ID

. Scaleform

Visual Studio

. Scaleform

Scaleform

Shipping

1.2 Scaleform 3.3 메모리 API 변경점

Scaleform 3.3

GfX::Movie

movie view

4

SysAlloc

"malloc

"

4K

512

SysAlloc

Scaleform::SysAlloc

Scaleform::SysAllocPaged

. SysAlloc

SysAllocPaged

. SysAllocStatic

2 메모리 할당

1.1.1 , Scaleform

- 1. SysAlloc
- 2. Scaleform
- 3. Scaleform

(1) (2)

Scaleform SysAlloc
Scaleform::SysAlloc::Alloc
Scaleform::SysAlloc::Free
Scaleform
(single shared global allocator)

SWF . SysAlloc
2.1

2.2

Scaleform

Scaleform 2

" "

2.1 할당 오버라이딩

Scaleform

1) SysAlloc . Alloc, Free, Realloc

2) Scaleform Gfx::System

malloc/free

Scaleform::SysAllocMalloc Scaleform

2.1.1 자신만의 SysAlloc 구현

Scaleform SysAllocMalloc

SysAllocMalloc

```
class MySysAlloc : public SysAlloc
{
public:
    virtual void* Alloc(UPInt size, UPInt align)
    {
        return _aligned_malloc(size, align);
    }

    virtual void Free(void* ptr, UPInt size, UPInt align)
    {
        SF_UNUSED2(size, align);
        _aligned_free(ptr);
        return true;
    }

    virtual void* Realloc(void* oldPtr, UPInt oldSize,
                          UPInt newSize, UPInt align)
    {
        SF_UNUSED(oldSize);
        return _aligned_realloc(oldPtr, newSize, align);
    }
};
```

```

        . MySysAlloc      SysAlloc
3      ( Alloc, Free, Realloc).
      , Scaleform      16      . oldSize      align
      Free/Realloc      .      , Alloc/Free
      wrapper      Realloc      .
      ,      Scaleform
Gfx::System
      .
      MySysAlloc myAlloc;
      GfxSystem gfxSystem(&myAlloc);

Scaleform 3.0      Gfx::System      Scaleform      ,
Scaleform      . Scaleform
      .      , Scaleform
      (Gfx::System      않는      ).
      Gfx::System::Init()      Gfx::System::Destroy()
      . Gfx::System      Gfx::System::Init()
SysAlloc      .

```

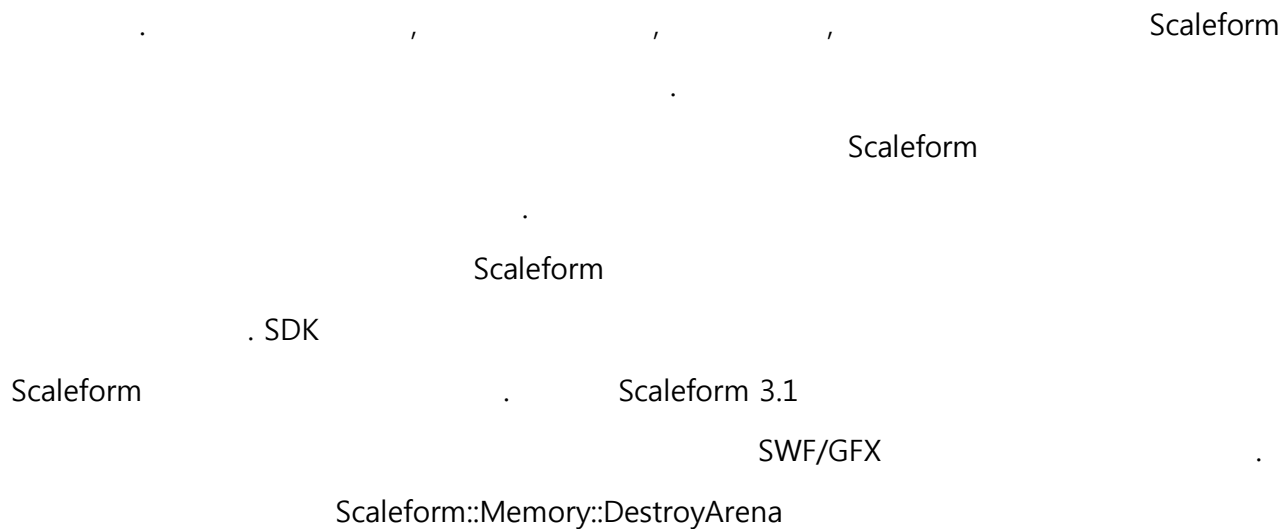
2.2 Scaleform 에 고정 메모리 블록 사용하기

```

      1
      , SysAlloc      Scaleform      .
      SysAllocStatic      .

void*      pmemChunk = ...;
SysAllocStatic      blockAlloc(pmemChunk, 6*1024*1024);
System      gfxSystem(&blockAlloc);
...

```

2.2.1.2 메모리 아레나의 사용

1. `Scaleform::Memory::CreateArena` `0(0` `|`
`)`
`SysAlloc` `SysAllocStatic`
2. `ID` `Gfx::Loader::CreateMovie/Gfx::MovieDef::CreateInstance`
`movie`

```

// Assume pbuf1 points to a buffer of 10,000,000 bytes.
SysAllocStatic sysAlloc(pbuf1, 10000000);
Memory::CreateArena(1, &sysAlloc);

```

```

// use arena 1 for the movie
Ptr<MovieDef> pMovieDef =

```

```

        *Loader.CreateMovie(filenameStr, Loader::LoadWaitFrame1, 1);
    ...
    // use arena 1 for the instance
    Ptr<Movie> pMovie =
        *pnewMovieDef->CreateInstance(MovieDef::MemoryParams(1), false);

3.          Gfx::MovieDef/Gfx::Movie
4.          . Gfx::MovieDef::WaitForLoadFinish
          . Gfx::ThreadedTaskManager

pMovieDef->WaitForLoadFinish(true);
pMovie = 0;
pMovieDef = 0;

( Scaleform::Ptr ):
pMovie->Release();
pMovieDef->Release();

5. DestroyArena . CreateArena
    . Scaleform::Memory::ArenaIsEmpty

// DestroyArena will ASSERT if memory was not properly released before
// the call. In Release it will crash at "return *(int*)0;".
Memory::DestroyArena(1);

SysAllocobject 'pbuff1' (scope)

```

2.3 OS

Scaleform OS

Scaleform::SysAllocWinAPI-VirtualAlloc()/VirtualFree() API . MS

Scaleform::SysAllocPS3 –PS3 sys_memory_allocate/sys_memory_free

OS

Scaleform WinAPI PS3 SysAlloc 64K

dlmalloc

2MB

2.3.1 SysAllocPaged 구현하기

Scaleform 3.3 SysAlloc , SysAlloc

SysAllocPaged . SysAllocPaged

Scaleform , SysAllocStatic OS

SysAlloc SysAllocPaged

SysAllocPagedMalloc

```
class MySysAlloc : public SysAllocPaged
{
public:
    virtual void GetInfo(Info* i) const
    {
        i->MinAlign    = 1;
        i->MaxAlign    = 1;
        i->Granularity  = 128*1024;
        i->HasRealloc   = false;
    }

    virtual void* Alloc(UPInt size, UPInt align)
    {
        // Ignore 'align' since reported MaxAlign is 1.
        return malloc(size);
    }
}
```

```

    }

    virtual bool Free(void* ptr, UInt size, UInt align)
    {
        // free() doesn't need size or alignment of the memory block, but
        // you can use it in your implementation if it makes things easier.
        free(ptr);
        return true;
    }
};

```

```

        . MySysAlloc      SysAllocPaged
    3      (GetInfo, Alloc, Free).      ReallocInPlace

    ,      .

GetInfo() –Scaleform::SysAllocPaged::Info

    .

Alloc –      .

    .      GetInfo      MaxAlign      . MaxAlign      1

    .

Free –Alloc      . free

    .

ReallocInPlace –      .

false      .

```

[Scaleform Reference Documentation](#)

```

    Alloc      Free      .

GetInfo      .      Scaleform      .

    SysAllocPaged      .      , Scaleform

    . SysAllocPaged::Info      4      .

MinAlign –

```

```

MaxAlign - 0 Scaleform
1 Alloc

Granularity -Scaleform . Scaleform
( malloc ) 64K

HasRealloc -ReallocInPlace false
SysDirectThreshold - null
SysDirectThreshold ,

MaxHeapGranularity - null , MaxHeapGranularity
alloc/free
MaxHeapGranularity
MaxHeapGranularity 4096 , 4096

```

```

MaxAlign SysAlloc
3

1.
MaxAlign 1 Scaleform
2.
MaxAlign 0
Alloc
3.
MinAlign MaxAlign
OS

```

Scaleform

Gfx::System

```
MySysAlloc myAlloc;  
Gfx::System gfxSystem(&myAlloc);
```

2.4

Scaleform

mesh

cache

Scaleform

API

2.4.1 힙 API

Scaleform

'new'

```
Ptr<FileOpener> opener = *new FileOpener();  
loader.SetFileOpener(opener);
```

'new'


```

        Gfx::SpriteDef      pHeap
    .
    Ptr<SpriteDef> def = *SF_HEAP_NEW(pHeap) SpriteDef(pdataDef);

    Gfx::SpriteDef      Scaleform      .      Scaleform
        Ptr<>
        . Scaleform::NewOverrideBase
        'delete'      . delete      SF_FREE(address)
    .

```

2.4.2 자동 힙

```

    .
    Scaleform::String/Scaleform::Array<>
    .
    ,
    .

```

```

class Sprite : public RefCountBase<Sprite>
{
    String      Name;
    Array<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();

```

```

Gfx::Sprite      ,
    .
    ,
    .
    Gfx::Sprite      ,
    .
    ,
    .

```

```

Scaleform
" "

SF_HEAP_AUTO_ALLOC(ptr, size)
SF_HEAP_AUTO_NEW(ptr) ClassName

SF_HEAP_ALLOC SF_HEAP_NEW

MemoryHeap* pheap = ...;
UByte * pbuffer = (UByte*)SF_HEAP_ALLOC(pheap, 100,
StatMV_ActionScript_Mem);
Ptr<Sprite> sprite = *SF_HEAP_AUTO_NEW(pbuffer + 5) Sprite();
...
sprite->DoSomething();
...
// Release sprite and free buffer memory.
sprite = 0;
SF_FREE(pbuffer);

, Gfx::Sprite
'pbuffer' SF_HEAP_AUTO_NEW
, Scaleform

Gfx::Sprite

class Sprite : public RefCountBase<Sprite>
{
    StringLH Name;
    ArrayLH<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);

```

```

        sprite->DoSomething();
        Scaleform::String  Scaleform::Array<>          "          (Local Heap)"
Scaleform::StringLH  Scaleform::ArrayLH<>          .
        "          "          .
        .
        Scaleform
        .

```

2.4.3 힙 추적기(Heap Tracer)

```

GfxConfig.h          SF_MEMORY_TRACE_ALL
Scaleform::Memory     SetTracer          CreateHeap, DestroyHeap, Alloc, Free
        .

```

```

SysAllocWinAPI
        .

```

```

#include "Kernel/HeapPT/HeapPT_SysAllocWinAPI.h"

class MyTracer : public Scaleform::MemoryHeap::HeapTracer
{
public:
    virtual void OnCreateHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnDestroyHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnAlloc(const MemoryHeap* heap, UPInt size, UPInt align,
                        unsigned sid, const void* ptr)
    {
        //...
    }

    virtual void OnRealloc(const MemoryHeap* heap, const void* oldPtr,
                        UPInt newSize, const void* newPtr)
    {
        //...
    }
}

```

```

        virtual void OnFree(const MemoryHeap* heap, const void* ptr)
        {
            //...
        }
};

static MyTracer tracer;
static SysAllocWinAPI sysAlloc;
static GFx::System* gfxSystem;

class MySystem : public Scaleform::System
{
public:
    MySystem() : Scaleform::System(&sysAlloc)
    {
        Scaleform::Memory::GetGlobalHeap()->SetTracer(&tracer);
    }
    ~MySystem() { }
};

SF_PLATFORM_SYSTEM_APP(FxPlayer, MySystem, FxPlayerApp)

```

3 메모리 보고

Scaleform Player AMP HUD

Scaleform 3.0

PC

Scaleform

AMP

AMP

AMP

Scaleform Player HUD UI , (Ctrl + F5)

. Scaleform::MemoryHeap::MemReport

```
void MemReport (class StringBuffer& buffer, MemReportType detailed,
               bool xmlFormat = false);
void MemReport (struct MemItem* rootItem, MemReportType detailed);
```

MemReport, XML

MemReportType

MemoryHeap::MemReportBrief – Scaleform

Memory 4,680K / 4,737K

Image	1,052
Sound	136
Movie View	1,739,784
Movie Data	300,156

MemoryHeap::MemReportFull – Ctrl+F6 Scaleform Player

SF_MEMORY_ENABLE_DEBUG_INFO	stat ID	memory
breakdown	.	.

Memory 4,651K / 4,707K

System Summary

System Memory FootPrint	4,832,440
System Memory Used Space	4,775,684
Debug Heaps Footprint	12,884
Debug Heaps Used Space	5,392

Summary

Image	1,052
Sound	136
Movie View	1,715,128
Movie Data	300,156

[Heap] Global	4,848,864
---------------	-----------

Heap Summary

Total Footprint	4,848,864
Local Footprint	2,427,860
Child Footprint	2,421,004
Child Heaps	4
Local Used Space	2,417,196
DebugInfo	120,832

Memory

MovieDef

Sounds	8
ActionOps	80,476
MD_Other	200

MovieDef

36

MovieView

MV_Other	32
----------	----

General

91,486

Image

92

Sound

136

String

31,425

Debug Memory

StatBag	16,384
---------	--------

Renderer

Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,160
TreeCache	13,220
TextureManager	2,336
MatrixPool	4,096
MatrixPoolHandles	2,032
Text	1,232

Renderer	13,488
Allocations Count	1,822

MemoryHeap:: MemReportFileSummary –

.	SWF	MovieData, MovieDef	MovieViews	stat ID
.	.	.	.	.
Movie File	3DGenerator_AS3.swf			
Memory			1,956,832	
MovieDef			152,664	
CharDefs			39,848	
ShapeData			1,716	
Tags			73,656	
MD_Other			37,444	
MovieView			196,249	
MovieClip			120	
ActionScript			181,917	
MV_Other			14,212	
General			1,596,712	
Image			960	
String			2,783	
Renderer			7,464	
Text			7,464	

MemoryHeap::MemReportHeapDetailed – AMP

Scaleform

Total Footprint	4,858,148
Used Space	4,793,972
Global Heap	2,405,492
MovieDef	80,720
Sounds	8
ActionOps	80,476
MD_Other	200
MovieView	32
MV_Other	32
General	92,566
Image	92
Sound	136
String	31,276
Debug Memory	8,192
StatBag	8,192
Renderer	2,182,960

Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,016
TreeCache	13,136
TextureManager	2,336
MatrixPool	8,192
MatrixPoolHandles	2,032
Text	1,232
Movie Data Heaps	300,156
MovieData "3DGenerator_AS3.swf"	300,156
MovieDef	152,664
CharDefs	39,848
ShapeData	1,716
Tags	73,656
MD_Other	37,444
General	141,332
Image	960
String	2,312
Movie View Heaps	1,727,936
MovieView "3DGenerator_AS3.swf"	1,727,936
MovieView	195,977
MovieClip	120
ActionScript	181,645
MV_Other	14,212
General	1,467,668
String	471
Renderer	7,464
Text	7,464
Other Heaps	360,580
_FMOD_Heap	360,580
General	359,944
Debug Data	12,884
Unused Space	51,292
Global	51,292
_FMOD_Heap	4,384
MovieData "3DGenerator_AS3.swf"	5,356
MovieView "3DGenerator_AS3.swf"	33,032

Scaleform SF_MEMORY_ENABLE_DEBUG_INFO

"stat ID"

MovieDef –

MovieView – Gfx::Movie

ActionScript

movie clip

CharDefs – movie clip flash

ShapeData –

Tags – keyframe

ActionOps – ActionScript

MeshCache – tessellator EdgeAA

FontCache –

4 가베지 콜렉션

Scaleform 2.0

2 (circular references)

Code:

```
var o1 = new Object;  
var o2 = new Object;  
o1.a = o2;  
o2.a = o1;
```

Scaleform 3.0

Scaleform 3.0

Scaleform

Include/GFxConfig.h

```
GFX_AS_ENABLE_GC  
// Enable garbage collection  
#define GFX_AS_ENABLE_GC
```

Scaleform

4.1

Scaleform 3.0

. Scaleform 3.3

, MovieView

MovieDef

```
Movie* MovieDef::CreateInstance(const MemoryParams& memParams,
                                bool initFirstFrame = true)
```

MemoryParams

```
struct MemoryParams
```

```
{
```

```
    MemoryHeap::HeapDesc    Desc;
```

```
    float                   HeapLimitMultiplier;
```

```
    unsigned                MaxCollectionRoots;
```

```
    unsigned                FramesBetweenCollections;
```

```
};
```

```
Desc - (MinAlign), (Granularity),
```

```
(Limit), (Flags) (Limit) , Scaleform
```

```
HeapLimitMultiplier - (0~1).
```

```
MaxCollectionRoots - roots , root
```

```
FramesBetweenCollections - roots
```

```
root
```

movie view

```
MemoryContext* MovieDef::CreateMemoryContext(
    const char* heapName,
    const MemoryParams& memParams,
```

```

                                bool debugHeap)

Movie* MovieDef::CreateInstance(MemoryContext* memContext,
                                bool initFirstFrame = true)

MemoryContext                    movie view

                                . movie view                    movie view
                                - (thread-safe)                    AMP
AMP
                                movie view
                                .

MemoryParams::Desc
                                . ( :ActionScript ) (Desc.Limit, HeapLimitMultiplier)
                                . 128K .(
.)
                                ,
                                . Boehm-Demers-Weiser(BDW)

BDW .( )
if (allocs since collect >= heap footprint * HeapLimitMultiplier)
    collect
else
    expand(heap footprint + overlimit + heap footprint *
                                HeapLimitMultiplier)

"collect"

HeapLimitMultiplier 0.25 . Scaleform
25%(HeapLimitMultiplier )
25%

Desc.Limit
Desc.Limit

```

```

        + Delta
    (
        ).

    MaxCollectionRoots/ FramesBetweenCollections.MaxCollectionRoots
        root . "root"
        ( ) root . ( : "obj.member=1" "obj" .)
    MaxCollectionRoots
        0
    FramesBetweenCollections
    "frame"
    FramesBetweenCollections 1800 30fps
    60 . DescLimit
    FramesBetweenCollections
    0 ,

```

4.2 Gfx::Movie:: ForceCollectGarbage

```

virtual void ForceCollectGarbage() = 0;

```